Project Specification

Unsupervised Learning Project | 1st Deliverable

1. Team Composition

Student Name	Student Number
Francisco Rodrigues dos Santos	72908
Duarte Ferreira	75419

2. Project Title

Hotel Booking Demand: Clustering of Booking Behaviour Profiles

3. Problem Statement

What is the project about?

We're applying clustering to the Hotel Booking Demand dataset to find out if there are real, meaningful groups in how people book hotels. 119,390 records, 32 features, mix of numerical and categorical. The pipeline goes from raw data validation through preprocessing to clustering, everything logged and reproducible. The main constraint is that we only use what was known at booking time, nothing that gets filled in afterward.

Why is this topic relevant or interesting?

Hotels don't come with pre-labelled customer segments, which is precisely why this is interesting as an unsupervised problem. The dataset has enough signal (lead times, prices, channels, room types, guest composition) that there is a real chance something meaningful is in there. The mixed-type, high-dimensional nature of the data after encoding also means the preprocessing choices genuinely affect results, which makes it a good testbed.

What kind of insight do you hope to obtain?

We're after profiles that are actually narratable, something a hotel manager could recognise, not just a partition that scores well on Silhouette. Early results are encouraging: iKMeans finds 3 clusters consistently with Silhouette around 0.60 and ARI = 1.0 across seeds. k-means is much messier (ARI as low as -0.01 on some runs), so the final answer depends on what Task 2 turns up.

4. Problem Framing (RQ1)

What exactly do you want to segment or group?

One row = one booking, and that is the unit. The question is: can we find distinct booking behaviour profiles using only what was known when the booking was confirmed? We call that the index time. Anything updated after that point (cancellation outcome, actual assigned room, waiting list position) is excluded from clustering inputs.

What would a useful segmentation help you understand?

Something more useful than the obvious splits like city vs resort or peak vs off-peak. We are thinking about profiles like early planners, last-minute solo travelers through OTAs, group bookings via corporate channels. That kind of structure could feed into pricing, communications and staffing. There is also a loyalty signal in there (is_repeated_guest, previous_bookings_not_canceled). It will be interesting to see whether repeat guests form their own cluster or spread across others.

Why is clustering appropriate here rather than another type of analysis?

There is no label to predict and no outcome variable that defines 'correct' segments. The point is to let the data show structure we don't already know about. Supervised methods need that label and we don't have one.

5. Representation / Preprocessing Choices (RQ2)

Included Variables

17 numerical, 10 categorical, 82 dimensions after OHE.

Numerical: lead_time, arrival date fields, stays_in_weekend_nights, stays_in_week_nights, adults, children, babies, is_repeated_guest, previous_cancellations, previous_bookings_not_canceled, booking_changes, days_in_waiting_list, adr, required_car_parking_spaces, total_of_special_requests.

Categorical: hotel, arrival_date_month, meal, country, market_segment, distribution_channel, reserved_room_type, assigned_room_type, deposit_type, customer_type.

Country has 93 unique values, anything below frequency 50 gets collapsed into "Other".
Top 15 countries cover 91% of bookings, so the tail does not carry much anyway.

Excluded Variables (Leakage Control)

Index time: booking confirmation. Leakage exclusions (not known at index time): is_canceled (booking outcome), reservation_status (post-arrival), reservation_status_date (date of that update).

Governance exclusions: ID-like, no behavioural content: agent (~13% missing), company (~94% missing).

All excluded columns can be used after clustering for profiling, just not as inputs.

Data Treatment

Numerical: median imputation then StandardScaler. Median because lead_time and adr have long right tails (EDA confirmed this). The IQR outlier check showed most "outliers" in integer columns are just zero-heavy distributions doing their thing, not errors. StandardScaler compresses them without needing to remove anything.

Categorical: fill missing with "Unknown", group rare categories at min_freq=50, then OneHotEncoder with handle_unknown="ignore". The grouper is mainly for country. handle_unknown="ignore" is there because the fast subsample might miss categories that appear in the full dataset. agent and company are dropped before the pipeline.

Goal Alignment

Without StandardScaler, lead_time in days would swamp adults just from the scale difference. That is the core reason for it. Median imputation makes sense given the skewed distributions.

The rare-category grouper exists because country without it turns into 90 mostly empty OHE columns and that inflates the space without helping separation.

From the EDA, arrival_date_year has a spurious negative correlation with arrival_date_week_number. It is a dataset window artifact: starts mid-2015, ends mid-2017, so early-year weeks only appear in later years and vice versa, which means it is not a real signal.

---

6. Limitations, Risks, or Ethical Issues

65 of our 82 features are binary OHE columns, and several are near-constant (deposit_type is almost always "No Deposit", customer_type mostly Transient). Euclidean distances in that kind of space are less reliable, since two bookings can look "far" apart because of a handful of categorical dummies even if their behaviour is similar. A PCA variant is planned to quantify how much this matters.

k-means instability is a real issue too. ARI of -0.01 on some seed combinations means you could get completely different clusters depending on initialisation. iKMeans is stable (ARI = 1.0) and will be the reference point.

country is included, so clusters could correlate with nationality. We are keeping it because it genuinely reflects booking patterns, but it is something to be explicit about when interpreting results. The dataset covers 2015-2017, pre-COVID. Booking behaviour has shifted enough since then that these results should not be generalised beyond that period.
